

Explicit Andrews–Curtis constructions for divisibility subfamilies of square presentations

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Abstract

Let u, y be free generators. For natural numbers p, s , define the ordered pair of relator words

$$G(p, s) = (u^2 y^{-(2s+1)}, uy^p u y^{-s} u^{-1} y^{-p}).$$

Setting both words equal to the identity defines the associated group presentation. We prove ordinary Andrews–Curtis triviality of this pair when $p \mid s$ or $p \mid (s+1)$; in particular, every $G(2, s)$ is covered. A uniform entry reaches a root-conjugacy pair. An iterated doubling identity then gives two arithmetic finishes, the second of which changes the root degree from p to $p-1$.

The quantified theorem is proved in Lean. Explicit products of conjugates also give an executable construction whose exported paths are checked by the official SAIR verifier. This construction can have exponential length; no shortness claim is made. For $G(3, 3m+1)$ we isolate an exact remaining companion-restoration condition. That condition and the full square-family problem are open.

1 Statement and context

Definition of the square family. Let u, y be the generators of the free group $F(u, y)$. For integers p, s , define

$$r_1(p, s) = u^2 y^{-(2s+1)}, \quad r_2(p, s) = uy^p u y^{-s} u^{-1} y^{-p}, \quad G(p, s) = (r_1(p, s), r_2(p, s)).$$

Here G is our notation for this specified ordered pair of words. A relator is a word imposed to equal the identity; imposing $r_1 = r_2 = 1$ gives the presentation

$$\langle u, y \mid u^2 = y^{2s+1}, \quad uy^p u = y^p u y^s \rangle.$$

We call it the square family because the first relation contains u^2 . The parameters p, s index the family; they are not additional generators. For example, $G(2, 1)$ has relations $u^2 = y^3$ and $uy^2 u = y^2 u y$. The AC problem concerns the specified relator pair, so throughout this paper $G(p, s)$ retains those words and their order, rather than denoting only the abstract group they present.

Allowed moves and solvability. Write $R \sim_{\text{AC}} S$ when one can pass from R to S by finitely many relator inversions, right multiplications by the other relator, and conjugations. The rank is always two. Put $\text{Sol}(R)$ for $R \sim_{\text{AC}} (u, y)$. Conjugation by an arbitrary word is counted as one mathematical primitive; the executable certificates expand it into conjugations by letters. Multiplication by the inverse of the other relator costs three primitives, including the two inversions that restore it.

Theorem 1.1 (Divisibility construction). *Let $p, s \in \mathbb{N}$, where $0 \in \mathbb{N}$. If $p \mid s$ or $p \mid (s + 1)$, then $\text{Sol}(G(p, s))$.*

The proof constructs the moves without assuming that the presented group is trivial. Thus triviality is also a consequence. The admissible case $p = 0$ in this theorem is precisely $s = 0$.

Corollary 1.2. *For every $s \in \mathbb{N}$, $\text{Sol}(G(2, s))$.*

Proof. One of $s, s + 1$ is divisible by two. □

Original coordinates and the square relation

The square relation arises from the original family

$$P(n, a, b, c) = (x^{-1}y^nxy^{-(n+1)}, \quad x^{-1}y^axy^bx^{-1}y^c). \quad (1)$$

For $n > 0$ this is a Miller–Schupp presentation with $w = y^axy^bx^{-1}y^c$. The following dictionary records two quotient-two specializations. A solution of the square pair in the right column gives a solution of the original pair in the left column:

Original tuple	Coordinate at $c = 0$	Square pair
$P(2s, s, p, c)$	$u = xy^{-p}$	$G(p, s)$
$P(2s + 1, p, s + 1, c)$	$u = xy^{-p}$	$G(p, s)$.

Here s, p are natural and c is integral. The transports, including the changes of basis, are witnessed by ordinary moves; their derivation is in Section 5.1. The first row at $s = 0$ is still a well-defined tuple even though the usual Miller–Schupp notation assumes $n > 0$.

Relation to earlier constructions

The relevant Miller–Schupp family is

$$\text{MS}(n, w) = \langle x, y \mid x^{-1}y^nx = y^{n+1}, \quad x = w \rangle,$$

where $n > 0$ and w has exponent sum zero on x . Shehper et al. prove AC-triviality for all $\text{MS}(1, w)$, for all $\text{MS}(n, y^{-1}xyx^{-1})$, and for $\text{MS}(2, y^{-k}x^{-1}yxy)$ for every integer k [1, Theorem B; Theorems 2, 3 and Corollary 9]. Their substitution operation precedes the witness operations used here. The $p = 1$ slice enters their $\text{MS}(1, w)$ class. Below we give the root-family witnesses directly.

Miasnikov–Myasnikov prove AC-triviality for balanced trivial-group presentations on two generators of total relator length at most twelve [3]. Lisitsa uses automated theorem proving to obtain simplifications of particular Miller–Schupp presentations and formulate parametric conjectures [4]. These provide finite classification and computer-assisted construction precedents for the present uniform argument. Bridson establishes lower bounds on AC complexity starting in rank four [5]. Our factor count measures one chosen rank-two construction; it does not bound the length of a shortest trivialization from below.

Substitution by conjugate products and transport under automorphisms are established techniques; see Panteleev–Ushakov [2]. Lackenby’s ordinary AC theorem for thickenable balanced presentations gives a different sufficient condition [6, Theorem 1.3]. We have not determined how the entire family here intersects that geometric class. Lean formalization of AC and automated certificate checking also have prior implementations [7, 9]. Our contribution is the arithmetic construction below and its formal proof. Priority for the underlying triviality classes has not been established by the comparison in this note.

2 Witnesses that compile to ordinary moves

Definition 2.1. A witness relative to a relator r , from a to b , is a finite ordered list $W = ((g_1, \epsilon_1), \dots, (g_t, \epsilon_t))$, with $\epsilon_i \in \{1, -1\}$, together with the identity in the ambient free group

$$a \prod_{i=1}^t g_i r^{\epsilon_i} g_i^{-1} = b. \quad (2)$$

We write $a \xrightarrow[r]{W} b$.

This contains more data than an equality in the presented quotient.

Lemma 2.2 (Witness compilation). *An identity (2) gives $(a, r) \sim_{\text{AC}} (b, r)$, with the second relator restored after each factor. A positive factor costs three mathematical primitives and a negative factor costs five. The analogous assertion holds with the two slots exchanged.*

Proof. For a positive factor, conjugate the donor r by g_i , right multiply the recipient, and conjugate the donor back by g_i^{-1} . For a negative factor, invert the donor before the multiplication and invert it again afterward. Repeat in the listed order. $\square$

The witness operations have explicit factor lists. Composition concatenates factor lists. Conjugation by h replaces every g_i by hg_i . Given $a \Rightarrow b$ and $c \Rightarrow d$ relative to the same donor, a witness $ac \Rightarrow bd$ is obtained by replacing each conjugator in the first list by $c^{-1}g_i$, then appending the second list. To obtain $a^{-1} \Rightarrow b^{-1}$, reverse the list, negate all signs, and replace each conjugator by ag_i . Iterating the product operation constructs positive powers; inverse words give negative powers. These rules follow directly by multiplying (2).

Three counts, and official work. For a witness with t_+ positive factors and t_- negative factors, the compiler used here has the following costs. In the last row, $|g_i|$ is the reduced letter length of the conjugator.

Witness factors	$t = t_+ + t_-$,
Mathematical primitives	$3t_+ + 5t_-$,
Official numbered moves	$\sum_{i=1}^t (2 g_i + 1) + 2t_-$.

These are costs of compiling the factors, before any entry or finishing moves. A numbered conjugation uses one generator letter; our negative-factor compiler retains the two donor inversions. Other official paths can use the numbered multiplication-by-inverse move directly. Identity conjugations still count as mathematical primitives but emit no numbered moves. For an exported path $R^{(0)}, \dots, R^{(L)}$, official work is $\sum_{j=0}^L (|R_0^{(j)}| + |R_1^{(j)}|)$, with each word freely reduced. The initial pair is included in

this sum. Thus 61 entry primitives, 81 primitives for a residual seed, and 183 or 21,819 numbered moves for complete examples measure different paths and different operations. None is a shortestness assertion.

3 Entry into a root-conjugacy pair

Set

$$B_p = y^{-p}u^2y^pu^{-1}, \quad A_s = y^{-(s+1)}u^{-1}y^su^{-1}. \quad (3)$$

Lemma 3.1 (Uniform entry). *For all integers p, s , a route of 61 mathematical primitives takes $G(p, s)$ to $(f(B_p), f(A_s))$, where*

$$f(u) = y^{s+1-p}u^{-1}y^p, \quad f(y) = y^{-1}.$$

We have $\text{Sol}(G(p, s))$ if and only if $\text{Sol}((B_p, A_s))$.

Proof. The complete route is recorded in the artifact described in Appendix A. After its first twelve macros, which cost eighteen primitives, the pair is

$$(y^{-p}u^{-1}y^pu^{-1}y^su, \quad uy^{-p}u^{-2}y^py^{-1}).$$

The remaining macros cost 43 primitives and end at the displayed image pair. Each step is a free-group identity, checked in the accompanying Lean files `ACDiagonalDescent` and `ACSquareP2Entry`.

The following identities explain the two donor substitutions in macros 13–51, in temporary bases related by the shears named below. They do not by themselves specify that route: the full schedule, including the normalizations and carried-back conjugators, is proved in `ACSquareP2Entry`. Temporary changes of basis and cyclic normalization put the first substitution in the following form:

$$\begin{aligned} R_0 &= u^{-2}y^{-p}uy^{-1}uy^p, & T_0 &= u^{-1}y^{-p}uy^su^{-1}y^p, \\ R_1 &= u^{-1}y^{-p}uy^{-(s+1)}uy^p. \end{aligned}$$

The relation T_0 replaces the second u^{-1} in R_0 by $y^{-p}uy^{-s}u^{-1}y^p$. The adjacent $u^{-1}u$ then cancels, leaving R_1 . More precisely,

$$R_0g_0T_0^{-1}g_0^{-1} = R_1, \quad g_0 = y^{-p}u^{-1}yu^{-1}y^pu.$$

A shear $u \mapsto uy^p$ and normalization arrange the second substitution:

$$\begin{aligned} R_2 &= u^{-2}y^{-p}uy^s, & T_2 &= u^{-1}y^{-p}uy^{p-s-1}uy^p, \\ R_3 &= u^{-1}y^{-p}u^{-1}y^{-p+2s+1}. \end{aligned}$$

Now replace the second u^{-1} in R_2 by $y^{-p}u^{-1}y^{-p+s+1}u^{-1}y^p$. The exact factor is

$$R_2g_2T_2^{-1}g_2^{-1} = R_3, \quad g_2 = y^{-s}u^{-1}y^pu.$$

After inverting the temporary generators, the shear $u \mapsto uy^{s+1-p}$ exposes the root pair, up to cyclic normalization. These changing bases explain the cancellations; they are not extra moves in the claimed route. The frozen schedule carries all conjugators back through the temporary bases and reaches the exact pair $(f(B_p), f(A_s))$.

To justify the solvability equivalence, the map f is an automorphism: its inverse sends u to $y^{-p}u^{-1}y^{p-s-1}$ and y to y^{-1} . A homomorphism sends every AC step to an AC step, with conjugators mapped by that homomorphism. If γ solves the root pair, the forward solution is

$$G(p, s) \xrightarrow{\text{entry}} f(B_p, A_s) \xrightarrow{f(\gamma)} (f(u), f(y)) \xrightarrow{\beta_f} (u, y).$$

Here β_f is an ordinary basis correction: conjugate its first relator by y^{p-s-1} , obtaining $u^{-1}y^{s+1}$; right multiply it by the second relator to power $s+1$, then invert both relators. For the converse, concatenate the reversed entry with a solution of $G(p, s)$, apply f^{-1} to that whole path, and correct its endpoint $(f^{-1}(u), f^{-1}(y))$. Conjugation of the first relator by y^p gives $u^{-1}y^{-s-1}$; right multiplication by the second relator to power $-s-1$, then two inversions, finishes. Each direction therefore transports a complete path and supplies its own endpoint correction. $\square$

4 The two arithmetic finishes

Lemma 4.1 (Iterated doubling). *For $p \in \mathbb{Z}$ and $m \in \mathbb{N}$, there is a witness relative to B_p*

$$y^{pm}uy^{-pm} \implies u^{2^m}.$$

The recursive list described here has $2^m - 1$ factors.

Proof. The basic witness from y^puy^{-p} to u^2 has the single positive factor with conjugator y^pu^{-1} ; explicitly,

$$(y^puy^{-p})(y^pu^{-1})B_p(y^pu^{-1})^{-1} = u^2.$$

Start at $m = 0$ with the empty witness. Write W_m for the stage- m list and put $d = 2^m$. The induction has two stages, both relative to B_p :

$$\begin{array}{ccc} y^{p(m+1)}uy^{-p(m+1)} & & \\ \Downarrow & W_m \text{ conjugated by } y^p & \\ y^pu^dy^{-p} = (y^puy^{-p})^d & & \\ \Downarrow & d \text{ copies of the basic witness} & \\ (u^2)^d = u^{2d}. & & \end{array}$$

In the second stage the product rule positions one copy at each of the d factors; the donor is restored after every copy. This is the source of the doubling growth. Conjugating a witness changes its conjugators but not its number of factors, so $L_0 = 0$ and $L_{m+1} = L_m + 2^m$. Summing this recurrence gives $L_m = 2^m - 1$. $\square$

Lemma 4.2 (Power finish). *For any integers k, e , $\text{Sol}((B_k, y^{-1}u^e))$.*

Proof. With donor $q = y^{-1}u^e$, the identity $yq = u^e$ supplies a one-factor witness replacing y by u^e . Apply its power and product witnesses to $B_k = y^{-k}u^2y^ku^{-1}$ to obtain u . The donor is restored. Right multiply the second relator by the first to power $-e$ to obtain y^{-1} , then invert. $\square$

Theorem 4.3 (Arithmetic root families). *For all $p \in \mathbb{Z}$ and $m \in \mathbb{N}$, both*

$$\text{Sol}((B_p, A_{pm})) \quad \text{and} \quad \text{Sol}((B_p, A_{pm+p-1}))$$

hold by finite witness constructions.

Proof. Put $d = 2^m$. The purpose in each case is to produce a companion $y^{-1}u^e$, while keeping a root relator available for the power finish.

Case 1: $s = pm$. Conjugate the companion by y^{pm} :

$$y^{pm} A_{pm} y^{-pm} = y^{-1} u^{-1} (y^{pm} u^{-1} y^{-pm}).$$

Use the inverse-word witness of Lemma 4.1 in the parenthesized factor. This leaves $y^{-1}u^{-(d+1)}$, to which Lemma 4.2 applies.

Case 2: $s = pm + p - 1$. The same conjugation first gives

$$y^{pm} A_{pm+p-1} y^{-pm} = y^{-p} u^{-1} y^{p-1} (y^{pm} u^{-1} y^{-pm}).$$

Applying the inverse doubling witness leaves the companion

$$q = y^{-p} u^{-1} y^{p-1} u^{-d}. \quad (4)$$

Set $v = u^{-1} y^{p-1} u^{-d}$. The exact identity $y^p q = v$ is a one-factor witness relative to q . Its inverse-word and product witnesses replace y^p and y^{-p} in $B_p = (y^p)^{-1} u^2 y^p u^{-1}$, while retaining q . The result is

$$\begin{aligned} v^{-1} u^2 v u^{-1} &= u^d y^{-(p-1)} u u^2 u^{-1} y^{p-1} u^{-d} u^{-1} \\ &= u^d y^{-(p-1)} u^2 y^{p-1} u^{-1} u^{-d} \\ &= u^d B_{p-1} u^{-d}. \end{aligned}$$

Only cancellation and commutation of powers of u were used in the middle line. Conjugate the first relator by u^{-d} to obtain (B_{p-1}, q) . Thus the old companion becomes the donor for changing the first relator; the resulting first relator becomes the donor for finishing the companion. Conjugate q by y^{p-1} and rewrite it as

$$y^{p-1} q y^{-(p-1)} = y^{-1} u^{-1} (y^{p-1} u^{-d} y^{-(p-1)}).$$

Since $(y^{p-1} u y^{-(p-1)})^{-d} = y^{p-1} u^{-d} y^{-(p-1)}$, the $(-d)$ -th power of the basic degree- $(p-1)$ doubling witness changes the parenthesized block to u^{-2d} . The companion is therefore $y^{-1} u^{-(2d+1)}$, as required for Lemma 4.2. At each substitution, Lemma 2.2 restores the donor, so the concatenation is a path of ordinary rank-two moves. $\square$

Proof of Theorem 1.1. If $s = pm$ with $m \in \mathbb{N}$, use the first case of Theorem 4.3 and Lemma 3.1. If $s + 1 = pk$ with $p, s, k \in \mathbb{N}$, then $k \geq 1$; set $m = k - 1$, so $s = pm + p - 1$, and use the second case. This includes the degenerate $p = s = 0$ through the first case. $\square$

The same argument also proves $\text{Sol}(G(p, pm))$ and $\text{Sol}(G(p, pm + p - 1))$ for arbitrary integer p and natural m . Natural divisibility in Theorem 1.1 is a convenient formulation with a nonnegative quotient. The doubling construction can produce long certificates; no short-path or polynomial-complexity claim follows from the theorem.

5 Parameter consequences and the remaining induction

We record the witnesses behind the parameter transports. To allow the two complementary heights to vary separately, put

$$S_N = u^2 y^{-N}, \quad T_{p,s} = u y^p u y^{-s} u^{-1} y^{-p}, \quad M_{p,s} = u y^p u y^{-p} u^{-1} y^{-s}.$$

Thus $G(p, s) = (S_{2s+1}, T_{p,s})$. Write $C_R(g) = g R g^{-1}$ for a conjugate of the donor R . In the original coordinates x, y , the standard pair in the notation Sol is (x, y) .

5.1 The original-coordinate transfers

Lemma 5.1 (Quotient-two pinches). *Let $n \in \mathbb{N}$, $a, b \in \mathbb{Z}$, and $q = x^{-1}y^a xy^b x^{-1}$. Then*

$$\begin{aligned} n = 2a &\implies P(n, a, b, 0) \sim_{\text{AC}} ((xy^{-b})^2 y^{-(n+1)}, q), \\ n + 1 = 2b &\implies P(n, a, b, 0) \sim_{\text{AC}} ((xy^{-a})^2 y^{-n}, q). \end{aligned}$$

Each pinch uses two negative donor factors; the second also uses inversion and conjugations of the first relator.

Proof. Set $a_0 = x^{-1}y^a x$ and $z = xy^{-b}$. Since $q = a_0 z^{-1}$,

$$a_0 \mathcal{C}_q(z^{-1})^{-1} = z.$$

When $n = 2a$, apply the square of this witness to $a_0^2 y^{-(n+1)}$. For the other pinch, use the one-factor identity

$$(xy^b x^{-1})q^{-1} = y^{-a}x.$$

If $n + 1 = 2b$, invert the first relator and conjugate it by x , exposing $(xy^b x^{-1})^2 y^{-n}$. The squared witness gives $(y^{-a}x)^2 y^{-n}$, and conjugation by y^a gives the stated endpoint. Lemma 2.2 restores q after each factor. These are the `left_pinch` and `right_pinch` paths in `ACSquareFamily`. $\square$

Lemma 5.2 (Mirror and sign transport). *For all integers N, p, s ,*

$$\text{Sol}((S_N, T_{p,s})) \iff \text{Sol}((S_N, M_{p,s})) \iff \text{Sol}((S_N, T_{-p,s})).$$

Proof. Let $\iota(u) = u^{-1}$ and $\iota(y) = y^{-1}$, and put $g = (uy^p u)^{-1}$. The free-group identities

$$y^{-N} S_N^{-1} y^N = \iota(S_N), \quad g M_{p,s}^{-1} g^{-1} = \iota(T_{p,s})$$

give a four-primitive path from $(S_N, M_{p,s})$ to $\iota(S_N, T_{p,s})$. Transport a complete solution by ι and invert both endpoint relators to correct (u^{-1}, y^{-1}) . Since ι^2 is the identity, this proves both directions of the first equivalence.

For the second equivalence, put

$$e = (uy^p)^{-1} M_{p,s}(uy^p) = uy^{-p} u^{-1} y^{-s} uy^p.$$

The two-factor identity

$$e \mathcal{C}_{S_N}(y^{-p} u^{-1} y^s) \mathcal{C}_{S_N}(y^{-p} u^{-1})^{-1} = T_{-p,s}$$

gives an ordinary path of nine primitives from the mirror pair to the negative-degree pair. Reversing this path completes the equivalence. The corresponding formal paths are `mirror_solvable` and `mirror_negative_steps`. $\square$

Lemma 5.3 (Complementary heights). *For integers p, s, k , put $N = s + k$. There is a thirteen-primitive path*

$$(S_N, T_{p,s}) \sim_{\text{AC}} (S_N, T_{p,k}).$$

Proof. Invert the second relator and conjugate it by u^{-1} , obtaining

$$e = u^{-1} T_{p,s}^{-1} u = u^{-1} y^p u y^s u^{-1} y^{-p}.$$

For $a = y^p u y^{-s}$, the required identity is

$$e \mathcal{C}_{S_N}(a u^{-1} y^{-p}) \mathcal{C}_{S_N}(a u^{-1})^{-1} \mathcal{C}_{S_N}(a) = T_{p,k}.$$

The three factors cost eleven primitives; the initial inversion and conjugation cost two. This is `complement_steps`. $\square$

Lemma 5.4 (Final-parameter shear). *For $n \in \mathbb{N}$ and integers a, b, c, t ,*

$$\text{Sol}(P(n, a, b, c)) \iff \text{Sol}(P(n, a, b, c - t)).$$

Proof. Let $\phi_t(x) = xy^t$ and $\phi_t(y) = y$. Conjugating both relators gives

$$y^t \phi_t(P(n, a, b, c)) y^{-t} = P(n, a, b, c - t),$$

where conjugation of a tuple acts on each component. Starting at the right-hand tuple, reverse those two conjugations, transport a solution of $P(n, a, b, c)$ by ϕ_t , and correct (xy^t, y) by right multiplying the first relator by the second to power $-t$. Reversing the shear gives the converse. Equivalently, the left shear $x \mapsto y^t x$ sends the original tuple directly to $P(n, a, b, c - t)$; the formal theorem `ACMillerSchupp.shear_family` uses this convention. $\square$

Deriving the dictionary. In the even row of Section 1, set $n = 2s, a = s, b = p$ in Lemma 5.1; in the odd row, set $n = 2s + 1, a = p, b = s + 1$. Both square relators become S_{2s+1} in the coordinate $u = xy^{-p}$. Inverting the donor q and conjugating it by x gives respectively $M_{p,s}$ and $T_{p,s+1}$ in that coordinate. Lemma 5.2 handles the mirror; Lemma 5.3 exchanges $s + 1$ and s . The coordinate map sends (u, y) to (xy^{-p}, y) , whose first relator is corrected by right multiplication by y^p . Finally, Lemma 5.4 supplies every integral c . Consequently the divisibility theorem applies to $P(2s, s, p, c)$ and $P(2s + 1, p, s + 1, c)$ whenever $p, s \in \mathbb{N}$ and $p \mid s$ or $p \mid (s + 1)$.

An independent group calculation. Every $G(p, s)$, for integral p, s , presents the trivial group. Indeed, in the presented group the second relation gives $y^{-p} u y^p = u y^s u^{-1}$. Squaring and using the first relation gives

$$u^2 = y^{-p} u^2 y^p = (u y^s u^{-1})^2 = u y^{2s} u^{-1}, \quad y^{2s} = u^2 = y^{2s+1}.$$

Thus $y = 1$, and the second relation then gives $u^2 = u$, so $u = 1$. This quotient-group calculation checks the defining relations; it supplies no additional ordinary AC path.

5.2 The period identity and covered residues

Lemma 5.5 (Four-factor period). *For integers N, p, s , the pair $(S_N, T_{p+N,s})$ reaches $(S_N, T_{p,s})$ in sixteen mathematical primitives, with the square fixed. In particular, for $s \in \mathbb{Z}$, solvability of $G(p, s)$ is periodic in p with period $2s + 1$.*

Proof. Set

$$\begin{aligned} g_1 &= y^{N+p} u y^s u^{-1} y^{-N}, & g_2 &= y^{N+p} u y^{s-N}, \\ g_3 &= y^{N+p} u y^{-N}, & g_4 &= y^p. \end{aligned}$$

The explicit identity is

$$T_{p+N,s} \mathcal{C}_{S_N}(g_1) \mathcal{C}_{S_N}(g_2)^{-1} \mathcal{C}_{S_N}(g_3) \mathcal{C}_{S_N}(g_4)^{-1} = T_{p,s}. \quad (5)$$

To derive it, first note the two-factor commutation witness, for any integer t :

$$(y^N u^t) \mathcal{C}_{S_N}(u^{-t} y^{-N}) \mathcal{C}_{S_N}(y^{-N})^{-1} = u^t y^N.$$

The first factor replaces y^N by u^2 ; the second replaces that square back by y^N after commuting it past u^t . Apply this with $t = 1$ and $t = -1$ to the middle word

$$y^N u y^{-s} u^{-1} \implies u y^{-s} y^N u^{-1} \implies u y^{-s} u^{-1} y^N.$$

Attach the prefix uy^p and suffix y^{-p-N} . The product rule in Section 2 gives exactly g_1, g_2, g_3, g_4 in that order. Two positive and two negative factors cost $3 + 5 + 3 + 5 = 16$ primitives. These are the factors of `squarePeriodWitness` in `ACSquareDiagonalFamilies`, with $k = N - s$. Iterating the path or its reverse proves the periodicity assertion. $\square$

Corollary 5.6 (Covered residues). *For $s \in \mathbb{N}$, $G(p, s)$ is AC-trivial whenever*

$$p \equiv 0, \pm 1, \pm 2, \pm s \pmod{2s+1}.$$

Proof. The cases $p = 1, 2, s$ follow from Theorem 1.1; signs and periods follow from Lemmas 5.2 and 5.5. For $p = 0$, conjugate the second relator by u^{-1} to obtain $q = uy^{-s}$. The witness $uC_q(u^{-1})^{-1} = y^s$ changes the first relator to y^{-1} . Invert the first relator, right multiply the second by the first to power s , and exchange the two slots to obtain (u, y) . $\square$

Corollary 5.7 (Heights three and four). *Both $G(p, 3)$ and $G(p, 4)$ are AC-trivial for every integer p .*

Proof. The residues $0, \pm 1, \pm 2, \pm 3$ exhaust the classes modulo seven. At height four, Corollary 5.6 covers all classes modulo nine except ± 3 . The proved seed $G(3, 4)$, its sign transport and its periods cover those two classes. $\square$

The seed is a finite kernel-checked certificate in `ACSquareP3Seed`; its 65 numbered moves start at the residual root pair associated with $G(3, 4)$, not at the square presentation itself. Its mathematical primitive cost is 81. Thus the height-four result retains a finite seed as well as the displayed uniform transports. By the dictionary, these two heights also give $P(6, 3, b, c)$, $P(7, a, 4, c)$, $P(8, 4, b, c)$ and $P(9, a, 5, c)$ for all remaining integral parameters. These are particular slices, not the full three-parameter families at those exponents.

5.3 What remains of exponent halving

Here is the precise limitation of the proposed extension. Put

$$Q_{r,d} = y^{-(r+1)}u^{-1}y^r u^{-d}.$$

Proposition 5.8 (Residual equivalence). *For $p, r \in \mathbb{Z}$ and $m \in \mathbb{N}$,*

$$\text{Sol}(G(p, pm + r)) \iff \text{Sol}((B_p, Q_{r,2^m})).$$

Proof. Use Lemma 3.1. Conjugating A_{pm+r} by y^{pm} gives $y^{-(r+1)}u^{-1}y^r(y^{pm}u^{-1}y^{-pm})$. The inverse-word doubling witness gives $Q_{r,2^m}$ while keeping B_p . All these ordinary paths can be reversed. $\square$

For the next residue $p = 3, r = 1$, a two-factor substitution and invertible coordinate transport give the equivalent pair (B_2, C_{2^m}) , where

$$C_d = y^{-1}u^{-1}y^{-1}u^{-1}yu^{-d}, \quad D_d = y^{-3}u^{-1}y^{-1}u^{-1}y^3u^{-d}.$$

Witnessed halving yields $(B_2, C_{2d}) \sim_{\text{AC}} (B_2, D_d)$, but the outer y -exponents have changed. Closing the induction requires

$$(B_2, D_d) \sim_{\text{AC}} (B_2, C_d), \quad d = 2, 4, 8, \dots \quad (6)$$

The formalization proves that (6) for all these d is equivalent to $\text{Sol}(G(3, 3m + 1))$ for every $m \in \mathbb{N}$, using the known $m = 0, 1$ seeds. Neither side of this equivalence is proved here. In particular, the first missing restoration is $d = 2$; halving alone does not establish it. This is a target for further work, and even its resolution would still leave other square-family parameters to treat.

6 Formal proof and executable evidence

The formalization uses Lean 4.29.1 and the pinned Mathlib revision in its Lake manifest. Its copy of the official `AC.lean` is byte-identical to the SAIR source [9] at commit

99a65377c5c4f412cd9af7b8d31c41464a855736.

The main statement is `AC.MillerSchupp.SquareFamily.P2.divisibility_square_solvable`. The root witness identity and both arithmetic finishes are quantified Lean declarations. Logical-dependency reports admit only the usual Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. The conjectures are definitions of propositions and are never assumed.

What is proved, and what is replayed. The Lean development proves the quantified AC constructions in this paper. Its witness identities, entry path, arithmetic finishes and basis corrections together establish the main declaration above in the official definition of ordinary AC reachability. The theorem does not depend on a Python execution or on the finite replay matrix.

The Python exporter is a separate implementation of the construction. It specializes the fixed affine route, constructs the finite witnesses, and expands them into the official ordinary rank-two move table. An independent word interpreter maintains intermediate states, and the unchanged SAIR verifier checks the final numbered path. The exporter also handles the admissible zero case explicitly. The finite CLI has resource bounds; those are not restrictions on the quantified Lean theorem. The exporter is not extracted from Lean, and no theorem identifies its complete factor lists with the Lean lists for every parameter. Agreement on finitely many parameter values would not prove that identification. An exported path is instead certified on its own by exact replay: each numbered step is applied to the input pair and the standard endpoint is checked. This gives reproducible finite certificates alongside the quantified formal proof, with distinct verification responsibilities.

The experiment uses 50 distinct pairs: the 44 distinct pairs (p, pm) and $(p, pm + p - 1)$ for $1 \leq p \leq 6$, $0 \leq m \leq 3$; all $(2, s)$ for $0 \leq s \leq 12$; and $(0, 0)$. All 50 pass unlimited official replay. Of these, 48 satisfy the golden budgets of 100,000 numbered moves, 10,000 peak total relator length, and 5,000,000 work units. The exceptions are $(2, 11)$ and $(2, 12)$, which exceed the work budget. The recorded construction time was 2.980 seconds and replay time 0.567 seconds on the machine identified in `receipts/replay.json`; these are one-run measurements, not portable performance guarantees.

p	s	Numbered moves	Official work	Peak length
2	0	183	3,185	35
2	3	282	9,215	83
2	6	637	58,022	273
2	10	5,457	2,766,856	1,273
2	11	19,176	14,902,075	1,779
2	12	21,819	26,190,511	2,829

All six overlapping search certificates replay independently and are shorter than the paths produced here. Those shorter certificates remain the appropriate choices for discovery-track scoring. The compared finite instances were already represented in the Caltech solved materials [8]; the present experiment validates a uniform proof construction. Certificate lengths, witness-factor counts and primitive costs are reported separately.

From the repository root, the reproduction commands are

```
python3 verify.py --lean
python3 experiment.py --matrix --output run/replay.json
```

The README gives dependency setup and reproduction instructions. Nine regression tests cover all numbered move types, donor restoration, the affine entry at signed parameters, witness corruption, boundary inputs, and equality with all 49 previously recorded constructed paths. The supplementary command `python3 paper/check_exposition.py` checks the new displayed entry factors, basis maps, coordinate pinches and degree-change identity on signed inputs. These are transcription checks, not additional proofs of the quantified statements.

Contributions and provenance. The research source is frozen at commit

68ceae3d422eee6de28bfd54793b21b2031cbda3.

The accompanying `SOURCE_PROVENANCE.json` records the copied and extracted source hashes. Robert Sneiderman directed the research and is the responsible author. GPT-6 Astra, an OpenAI model used through Codex, contributed to mathematical exploration, witness construction, Lean and Python development, and manuscript preparation. Fable’s analysis identified the quotient-two pinch and the $p = \pm 1$ connection to the published $MS(1, w)$ theorem. Further source and contribution details are recorded in the repository. In particular, the source history groups the uniform entry, doubling and $B_p \rightarrow B_{p-1}$ finish in the same proof-development pass; the later halving pass introduces C_d, D_d and the exact restoration condition. Both are Codex-assisted developments under Sneiderman’s direction. The records do not establish a separate discovery attribution for each identity or an exact model version for every earlier session.

A The affine schedule as a reproducibility artifact

The complete schedule is distributed as `entry_route.json`; `paper/entry-table.tex` is its generated typeset listing. In that listing, slots are indexed by $0, 1$. I_i inverts relator i , M_i^ϵ replaces it by $R_i R_{1-i}^\epsilon$, and $C_i(g)$ conjugates it by g . An M_i^{-1} entry expands into inversion of the donor, multiplication, and inversion of the donor again. Apply the rows in order to $G(p, s)$. Rows 1–12 reach the normal form in Lemma 3.1; rows 13–51 reach $f(B_p, A_s)$. Thus the 51 macros comprise 61 primitives. Section 3 explains the two donor substitutions and the automorphism transport; the schedule fixes every intervening normalization. `python3 paper/generate_table.py` regenerates the listing. The corresponding affine identities are proved for all integer p, s in the pinned Lean source. The package manifest records hashes for the JSON, generated listing and formal source.

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